



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.7

Generate Equivalent Expressions

Lesson 6-8 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can use the commutative, associative, and identity properties to rewrite expressions.

Language Objective: I can explain my reasoning using the words commutative property, associative property, and identity property.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Properties of Operations

Commutative Property

Associative Property

Identity Property

Property

Distributive Property

Equivalent expressions

Expand

Factor an expression



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Rules that let you reorder or regroup without changing the

.

2

The rule that order does not change a sum or product, like $a + b = b + a$, is the

.

3

The rule that grouping does not change a sum or product, like $(a + b) + c = a + (b + c)$, is the .

4

The rule that adding 0 or multiplying by 1 keeps a number the same is the

.

5 A rule that is always true in math is a .

6 The rule that lets you multiply a number across a sum, like $a(b + c) = a \times b + a \times c$, is the .

7 Expressions that always have the same value are .

8 Writing $6(3x + 4)$ as $18x + 24$ is called the expression.

9 Writing $18x + 24$ as $6(3x + 4)$ is called .

Write About the Math

What properties did the engineer use, and why does rearranging make the calculation easier?

¿Qué propiedades usó la ingeniera, y por qué reacomodar hace más fácil el cálculo?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Properties of Operations**

Propiedades de las operaciones

☐ **Commutative Property**

Propiedad conmutativa

☐ **Associative Property**

Propiedad asociativa

☐ **Identity Property**

Propiedad de identidad

☐ **Property**

Propiedad

Model: *The commutative property lets you change the order of a sum without changing the total. — Students name the commutative property and explain that $3 + 5 + 2$ gives 10 in any order because reordering addends does not change the sum.*

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The engineer used the ____ property to change the ____ and the ____ property to change the ____.** This makes it easier because $4 \times 25 = \underline{\hspace{1cm}}$, which is a friendly number to multiply by.
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**

Mi evidencia es _____, entonces _____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Properties of Operations
2. **Notes 2:** Commutative Property
3. **Notes 3:** Associative Property
4. **Notes 4:** Identity Property
5. **Notes 5:** Property
6. **Notes 6:** Distributive Property
7. **Notes 7:** Equivalent expressions
8. **Notes 8:** Expand
9. **Notes 9:** Factor an expression
10. **Watch for this mistake:** *A common mistake in Properties of Operations is applying the commutative property where it does not belong, or confusing it with the associative property. Students often assume $15 - 8 = 8 - 15$ (it does not: $7 \neq -7$ — commutative only works for addition and multiplication), or they see parentheses move and call it "associative" even when the numbers inside just swapped places, like $(2 + 3) + 4 = (3 + 2) + 4$ (that is commutative — the order inside the parentheses changed, not the grouping). Before you label a property, ask: did only the order change (commutative), did only the grouping change (associative), or was a 0 or 1 involved (identity)?*
11. **Try It 1:** C. Associative Property — *The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.*
12. **Try It 2:** D. Identity Property — *Multiplying by 1 keeps the same value, so this is the Identity Property of Multiplication.*